

Metadata for SPB annual summaries

20-Jan-23

Total of 23,664 records, each for one county, borough, or Federal administrative unit in one year.

Includes 3775 records with SPB trapping (*SPB.trapping* = 1).

Trapping data came from keypunching values from the original data sheets that were mailed to Ron Billings. All data were keypunched twice to check for keypunching errors.

Includes 18,551 records with spot data (*Spots0.data* = 1)

Includes 3612 records for which predictions were possible (*Valid-for-prediction* = 1)

Includes 3339 records with predictions and records of resulting spots (*Predictions-and-outcome* = 1)

Includes 19,889 records with spot data but no trapping data (*SPB.trapping* = 0 and *Spots.data* = 1).

Compared to what was available for download from spbpredict.com in January 2022, the data frame includes 19,889 more records with spot data but no trapping data. There were also some corrections, most notably to *Clerids.1*.

This data frame has passed QAQC by Matt Ayres in January 2022.

For all columns, data that are missing or not applicable are represented by an empty field.

ColumnName	Description
<i>State</i>	State
<i>State.2</i>	State abbreviation
<i>County-or-Federal</i>	County or Federal administrative unit
<i>County</i>	County or Parish name if applicable or "mixture" for federal lands (usually Ranger Districts) that cross boundaries of counties or parishes.
<i>FIPS</i>	FIPS state and county codes
<i>FederalName.2020</i>	Name of federal land as in Survey123 as of 2020
<i>FederalName.old</i>	Name of federal land that has historically been used in SPB southwide trapping program (usually a Ranger District).
<i>FederalName.older</i>	An older name of federal land that has historically been used in SPB southwide trapping program (usually a Ranger District).
<i>Year</i>	Year of trapping
<i>SPB.trapping</i>	Was there SPB trapping in this place in this year? 0 = no. 1 = yes.
<i>Valid-for-prediction</i>	Is there basis for a prediction? 0 = no. 1 = yes. ("Yes" requires that <i>SPB.trapping</i> = 1, and that <i>spots.1</i> and <i>spots.2</i> are numbers).
<i>Spots0.data</i>	Were there spot counts for this year in this place? 0 = no. 1 = yes.
<i>Prediction-and-outcome</i>	Were there both predictions and spot data for this year in this location? 0 = no, 1 = yes (requires that <i>Valid-for-prediction</i> = 1 and <i>Spots0.data</i> = 1)
<i>Endobrevicomin</i>	Endobrevicomin included in trapping? 0 = no. 1 = yes. All no during 1988 - 2009
<i>totalTrappingDays</i>	Total number of trap days for this county or forest in this year
<i>trapCount</i>	Total number of traps for this county or forest in this year
<i>days-per-trap</i>	Average number of days / trap (<i>totalTrappingDays</i> / <i>trapCount</i>)
<i>TotalSPB</i>	Total SPB captured in this county or forest in this year (= <i>SPB.14days.orig</i> / 14 · <i>totalTrappingDays</i>).
<i>SPB.14days.orig</i>	Number of SPB captured per trap per 14 days.
<i>SPB.14days.corrected</i>	SPB.14days.orig corrected for years when no endobrevicomin was used. Then, orig x 10 (per Brian Sullivan)
<i>Clerids.14days</i>	Number of clerids captured per trap per 14 days. NA = not available.
<i>Clerids.1</i>	Number of clerids captured per trap per 14 years in this location last year.
<i>Spots.0</i>	Number of local infestations of southern pine beetles ("spots") in year of trapping within county, parish, or ranger district
<i>Spots.1</i>	Number of local infestations of southern pine beetles ("spots") in year previous to trapping within county, parish, or ranger district
<i>Spots.2</i>	Number of local infestations of southern pine beetles ("spots") two years previous to trapping within county, parish, or ranger district
<i>pi</i>	Parameter estimated by zero inflated poisson model
<i>mu</i>	Parameter estimated by zero inflated poisson model
<i>expSpots_if_spots</i>	Expected number of spots if there are spots
<i>ProbSpots.GT.0</i>	Predicted probability of any beetle spots in the year of trapping within the county, parish, or ranger district.
<i>ProbSpots.GT.20</i>	Estimated probability of > 20 spots
<i>ProbSpots.GT.50</i>	Estimated probability of > 50 spots
<i>ProbSpots.GT.150</i>	Estimated probability of > 150 spots
<i>ProbSpots.GT.400</i>	Estimated probability of > 400 spots
<i>ProbSpots.GT.1000</i>	Estimated probability of > 1000 spots
<i>ln(SPBCorrected+1)</i>	Natural logarithm of (SPB.14days.corrected + 1). Used for averaging trap captures across locations or years (because trap captures have a log-normal distribution)
<i>ln(clerids+1)</i>	Natural logarithm of (Clerids.14days + 1). Used for averaging trap captures across locations or years (because trap captures have a log-normal distribution)
<i>ln(spots+1)</i>	Natural logarithm of (Spots.0 + 1). Used for averaging spot numbers across locations or years (because spots have a log-normal distribution)
<i>logit(Prob>50)</i>	logit transformation of ProbSpots.GT.50. Used for averaging of predicted probabilities across locations or years. Calculated as $\ln(P / (1-P))$ where $P = \text{ProbSpots.GT.50}$. [Equation for back-transformation: $P = e^{(\text{logit})} / (1 + e^{(\text{logit})})$]
<i>pred.Spots.logUnits</i>	Predicted number of spots in units of natural logarithm. Calculated as $0.129 + 6.743 (\text{ProbSpots.GT.50})$. Based on linear regression of $\ln(\text{spots}+1)$ vs. ProbSpots.GT.50 for 3522 observations from 1988 - 2000.
<i>pred.Spots.origUnits</i>	Back-transformation of predicted spots. Calculated as $e^{(\text{Pred.Spots.logUnits})-1}$
<i>residualSpots.logUnits</i>	Difference between observed and predicted number of spots. Calculated as $\ln(\text{spots}+1) - \text{pred.Spots.logUnits}$